

Where does the Sun rise?

A general equation for the Sun's azimuth — built on NOAA's verified solar-position equations, in the views of Class-XII student, and applied to a 22 km region around 23.169753993380713° N, 82.36029982215409° E

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Abstract

Two questions any student can ask — *Where does the Sun rise?* and *where exactly is the Sun right now, and how can I calculate it?* — have exact answers that fit inside Class-XII mathematics. We take the solar-position equations published by the U.S. National Oceanic and Atmospheric Administration (NOAA), a verified source whose formulae are quoted here verbatim with their provenance, and reduce them with nothing more than the syllabus tools of trigonometry, inverse trigonometric functions and one spherical-triangle identity. The reduction produces a single *general equation* for the Sun's azimuth,

$$A = 180^\circ + \tan^{-1}[\sin H / \cos H \sin \phi - \tan \delta \cos \phi],$$

which needs no separate morning/afternoon cases and is valid at every hour, in every season. Setting the zenith distance to the standard sunrise value $z_0 = 90.833^\circ$ gives the classroom sunrise formula $A_0 = \cos^{-1}[(\sin \delta - \sin \phi \cos z_0) / (\cos \phi \sin z_0)]$. Every formula is validated against NOAA's own operational algorithm, which we execute exactly as published: our transcription agrees to machine precision, and the simplified model reproduces the sunrise azimuth for all 365 days of 2025 with a maximum error of 1.16° (rms 0.57°) and sunrise times to within 2.0 minutes. An independent check against a public ephemeris service agrees to 0.14° , and reproduces its live Sun direction at the time of writing. We then answer the two opening questions for a 22 km radius region centred on 23.169753993380713° N, 82.36029982215409° E (Duman Hill, near Chirimiri, Khadganva tahsil, Manendragarh–Chirimiri–Bharatpur district, Chhattisgarh): the Sun rises between 64.0° (21 June) and 115.2° (21 December) — a swing of 51.3° — and within the entire 22 km disc the rising direction is constant to better than 0.09° and the rising clock time to within 1.9 minutes, so one table serves the whole district. Nothing new is

invented: every equation descends, step by step, from a published source, and each step is checked numerically.

Keywords: solar azimuth; solar declination; equation of time; hour angle; spherical astronomy; sunrise; equation of time; Indian Standard Time; CBSE Class XII; NCERT; Chhattisgarh.

Introduction

Two questions, one small box of mathematics

Everyone who has watched a sunrise knows that the Sun does not always come up in the same place. Standing at the same spot every morning for a year, a careful observer sees the rising point slide northward along the horizon until about 21 June, then return, then slide southward until about 21 December, and then return again. The two everyday questions

(i) *Where does the Sun rise?* (ii) *Exactly where is the Sun right now, and how do I calculate it?*

therefore have answers that change every single day. At the site studied in this paper the change is large: the Sun rises 51.3° further north on 21 June than it does on 21 December. Neither answer requires a telescope, a computer, or a new theory — both are fixed by three angles and the trigonometry that is already in the Class-XI and Class-XII NCERT syllabus.

The purpose of this paper is to show, in full and with numerical proof, how to get from a published, professionally verified set of equations to a formula a Class-XII student can use with a scientific calculator, and to apply that formula to one specific region of India. We stress at the outset, as the title of the project demands:

We are not inventing anything. Every equation in this paper is either (a) quoted from NOAA's published solar-position sheet and its published computer code, or (b) obtained from those equations by algebraic steps that a Class-XII student can follow. The contribution of this paper is the *reduction*, the *rearrangement* into one general equation, the *verification*, and the *local application* — not new astronomy.

Why azimuth is the right quantity to study

The direction of any object in the sky is fixed by two angles (Fig. Fig. 1): its *altitude* a , measured upward from the horizon, and its *azimuth* A , measured *clockwise from true North*, so that $A = 0^\circ$ is North, 90° East, 180° South and 270° West. Altitude answers “how high”, azimuth answers “in which direction”. When the Sun is on the horizon, its altitude is zero and the azimuth alone answers the question “where does the Sun rise?”. At every other moment of the day the pair (a, A) locates the Sun exactly, so that — as we show in Section 4 — a single formula for A answers question (ii) for all 365 days and all 24 hours.

This has immediate practical uses that a student can defend in a project viva: planning the orientation of solar panels, predicting whether a shadow will fall on a particular wall, fixing the direction of a new building, understanding why temple doorways in India were traditionally aligned east, or simply winning an argument about whether the Sun really rises “in the east”.

Approach: three levels, one sky

The same sky is described here at three levels of mathematics (Table Table 1). All three are compared against each other in Section 7, so a reader can see exactly what is lost when a formula is simplified, and what is not.

The three levels of description used in this paper.

Level	Model	Tools needed	Role in this paper
2	NOAA's full algorithm (Refs. [1, 2])	$+$, $-$, $\times$, $\div$, $\sin$, $\cos$, $\cos^{-1}$	the reference standard
1	NOAA's printed equation sheet (Ref. [1])	the same, fewer series terms	the "verified source" formulas
0	the Class-XII model of Section 4	one $\sin$, one $\cos$, one $\tan^{-1}$	what a student can use by hand

What is in this paper

Section 2 fixes the vocabulary and derives the one geometric identity everything rests on. Section 3 quotes the verified source. Section 4 performs the reduction and produces the general equation, Eq. (22). Section 5 specialises it to sunrise and sunset and explains three facts a student can check on the ground. Section 6 is the case study: the 22 km region around the given coordinates, with annual tables, an hourly table, a by-hand worked example, the June near-zenith passage peculiar to this latitude, and a compass experiment. Section 7 verifies every claim numerically. Section 8 discusses accuracy, limits and extensions, and Section 9 gives ten classroom exercises with answers. Appendices contain NOAA's own algorithm as published, a complete student program, and the algebra that connects our general equation to NOAA's printed one.

Mapping to the NCERT syllabus

Table Table 2 shows that the paper uses no mathematics beyond what a Class-XII student has already met. The physics and geography it leans on come from books the student has already read: the class-six chapter on the motions of the Earth (rotation, revolution, the 23.44° tilt, solstices and equinoxes), the class-eleven chapters on solar radiation and on latitude, longitude and time, and the class-eleven and class-twelve mathematics chapters on trigonometric and inverse-trigonometric functions.

Where each idea comes from in the NCERT syllabus.

NCERT chapter	Concept used here	Used in
Geography, Class VI, <i>Motions of the Earth</i> (Ch. 3)	rotation (15° per hour), revolution, axial tilt 23.44° , solstice, equinox	Section 2, Section 4, Section 5
Geography, Class XI, <i>Latitude, Longitude and Time</i> (Practical Work, Ch. 3)	latitude and longitude, local vs standard time, 4 minutes per degree, IST meridian 82.5° E	Section 4, Section 6
Geography, Class XI, <i>Solar Radiation, Heat Balance and Temperature</i>	solar radiation, effect of latitude, angle of incidence	Section 2, Section 6
Physics, Class XI (units, measurement), Class XII (ray optics: refraction)	angle measurement, uncertainty, atmospheric refraction of light	Section 3, Section 8

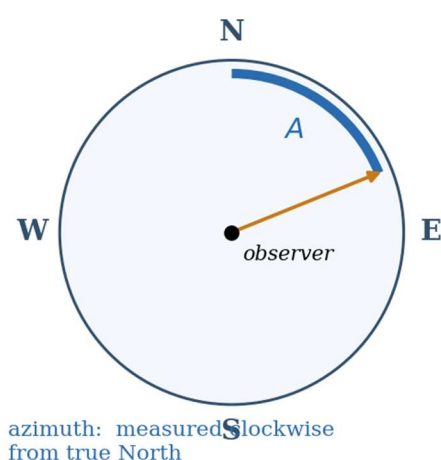
NCERT chapter	Concept used here	Used in
Mathematics, Class XI, <i>Trigonometric Functions</i> (Ch. 3)	sine and cosine rules, radians, periodic functions	Section 2, Section 5
Mathematics, Class XII, <i>Inverse Trigonometric Functions</i> (Ch. 2)	principal values of $\sin^{-1}$ and $\tan^{-1}$, range $(-\pi/2, \pi/2)$ of $\tan^{-1}$	Section 4

The sky in coordinates a student can draw

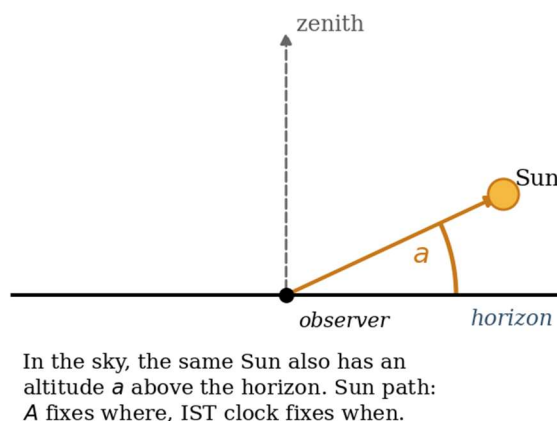
Azimuth and altitude

Figure Fig. 1 shows the observer's local sky in the two views a student should be able to sketch on a blackboard: looking straight down (plan view) and looking sideways (elevation view).

(a) looking straight down: azimuth A



(b) looking sideways: altitude a



The two angles that fix the Sun. (a) Looking down from the zenith: the azimuth A is the angle from true North, measured clockwise through East. (b) Looking sideways: the altitude a is the angle between the Sun's direction and the horizon.

The azimuth answers "where?", the altitude answers "how high?".

The horizon circle in the plan view is the *celestial horizon* — the 360° of directions available at eye level. The point straight overhead is the *zenith*, and the angle between the zenith and the Sun is the *zenith distance* z . Altitude and zenith distance are complementary:

$$z + a = 90^\circ.$$

Two conventions are worth stating because examination questions are full of them. First, azimuth is measured from *true* North (the direction of the Earth's rotation axis), not from magnetic North, and not from South; some older texts use South-based azimuths, which differ by 180° . Second, altitudes are counted positive above the horizon and negative below it. A compass measures magnetic North; at the site studied here the difference between true and magnetic North happens to be nearly zero (Section 6.7), but at other places it can exceed 20° , which is why the distinction matters.

The Sun's three coordinates: latitude, declination, hour angle

An observer has a fixed coordinate, the *latitude* ϕ (positive north of the equator). The Sun has two coordinates that change with time:

- the **declination** δ — the Sun's “latitude in the sky”, i.e. the latitude on Earth where the Sun is exactly overhead at that moment. It swings between $+23.44^\circ$ (around 21 June, summer solstice) and -23.44° (around 21 December, winter solstice) and is zero at the equinoxes. This single number, familiar from the class-six chapter on the motions of the Earth, is what makes the sunrise point move along the horizon.
- the **hour angle** H — how far the sky has turned since the Sun crossed the observer's meridian. The Earth rotates 360° in 24 h, that is, 15° per hour, so

$$H = 15^\circ \times (\text{local solar time in hours} - 12).$$

H is negative before local noon, zero at local noon, and positive after it. It is the “what time is it” coordinate of the Sun.

Latitude ϕ , declination δ and hour angle H together fix the Sun completely; the pair (ϕ, δ) fixes the sunrise point. All the symbols used in this paper are collected in Table Table 3.

Symbols used in this paper. Angles are in degrees unless a formula says otherwise.

Symbol	Meaning	Typical value at the study site
ϕ	latitude of the observer (north positive)	23.169754° N
λ	longitude of the observer (east positive)	82.360300° E
λ_s	standard meridian of the local time zone (IST)	82.5° E
T_z	time-zone offset from UT	+5.5 h
δ	solar declination	-23.44° to $+23.44^\circ$
H	hour angle of the Sun	-180° to $+180^\circ$
H_0	hour angle at sunrise (its magnitude)	80.3° (21 Dec) to 101.7° (21 Jun)
A	azimuth, clockwise from true North	0° to 360°
A_0	azimuth at sunrise (“the rising point”)	64.0° to 115.2°
a, z	altitude and zenith distance, $z + a = 90^\circ$	$z = 90.833^\circ$ at sunrise
N	day number of the year, 1 January = 1	—
δ_0	declination used in the simplified model	—
EoT	equation of time (clock model minus sundial)	-14.2 to $+16.4$ min
z_0	standard zenith distance of sunrise	90.833°

The astronomical triangle: the whole theory in one picture

Every formula in this paper comes from one triangle drawn on the celestial sphere (Fig. Fig. 2). Its three corners are

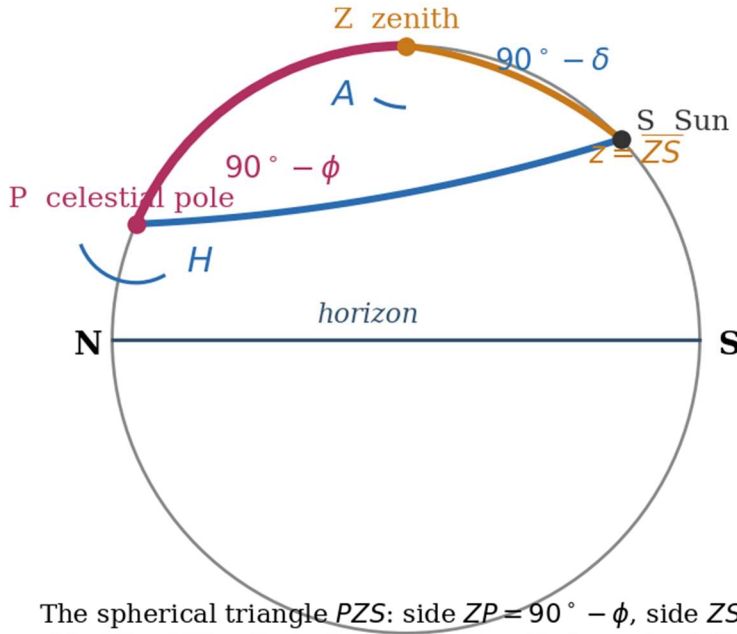
- Z , the observer's **zenith**;
- P , the **celestial pole** — the point of the sky above the Earth's north pole, which is fixed at altitude ϕ in the north for an observer at latitude ϕ ; and

- S , the Sun.

The three sides of this triangle are known from the definitions above:

$$ZP = 90^\circ - \phi, \quad ZS = z \text{ (the zenith distance)}, \quad PS = 90^\circ - \delta,$$

and two of its angles are known: the angle at Z is the azimuth A (with the convention of Fig. Fig. 1), and the angle at P is the hour angle H . Everything else — the Sun's altitude, and therefore its shadow, and its azimuth — follows from solving this triangle. This is the same triangle that every navigation textbook draws, and it is the reason a sailor can find a ship's position from the Sun.



The spherical triangle PZS : side $ZP = 90^\circ - \phi$, side $ZS = z$, side $PS = 90^\circ - \delta$, angle at $Z =$ azimuth A , angle at $P =$ hour angle H . The two cosine rules applied to it give every formula in this paper.

The astronomical triangle PZS on the celestial sphere, seen from outside as the observer's meridian plane lies in the page. The Sun S is drawn off the meridian. Side $ZP = 90^\circ - \phi$, side $PS = 90^\circ - \delta$ and side $ZS = z$; the angle at Z is the azimuth A , the angle at P is the hour angle H . Two applications of the spherical cosine rule to this one triangle produce every equation in the paper.

For a spherical triangle with sides a, b, c (in degrees of angular measure) and the angles A, B, C opposite them, the cosine rule takes two forms:

$$\cos a = \cos b \cos c + \sin b \sin c \cos A \quad (\text{for the sides}), \quad (1)$$

$$\cos A = \frac{\cos a - \cos b \cos c}{\sin b \sin c} \quad (\text{for an angle}). \quad (2)$$

Equation (1) lets us find a side from two sides and the included angle; Eq. (2) lets us find an angle from the three sides. Between them, these two lines are the entire mathematical machinery of this paper. Applying Eq. (1) to the side ZS ($a \rightarrow z$, $A \rightarrow H$, $b \rightarrow 90^\circ - \phi$, $c \rightarrow 90^\circ - \delta$) gives, since $\cos(90^\circ - x) = \sin x$,

$$\cos z = \sin \phi \sin \delta + \cos \phi \cos \delta \cos H. \quad (3)$$

Equation (3) is the standard relation between the Sun's zenith distance and its hour angle; together with $H = 15^\circ (t - 12)$ it answers “how high is the Sun at clock time t ?”. It is quoted, in exactly this form, in NOAA's published sheet (Ref. [1]) that we take up in the next section.

The verified source: NOAA's published solar-position equations

Provenance

The equations used in this paper are taken from the U.S. National Oceanic and Atmospheric Administration (NOAA), Global Monitoring Laboratory, which publishes both a one-page equation sheet, *General Solar Position Calculations* (Ref. [1]), and the source code of its public Solar Calculator (Ref. [2]). These formulae are not a classroom simplification: they are the operational equations behind a national agency's calculator, widely used by engineers, architects, surveyors and solar-energy practitioners, and they are quoted here verbatim (with one change of symbol, explained below) so that a reader can check the transcription. We use two verified sources in this paper:

1. the printed sheet (Ref. [1]) — closed-form expressions for the declination, the equation of time, the hour angle, the zenith distance and the azimuth (this is “Level 1” of Table Table 1); and
2. the published JavaScript implementation (Ref. [2]) — the same quantities computed from an accurate solar ephemeris (this is “Level 2”, our reference standard).

In addition, we compare with an independent public ephemeris service (Ref. [4]) in Section 7.4.

The equations as published

NOAA begins with the *fractional year* γ , a number that turns the calendar date into the Sun's position along its orbit:

$$\gamma = \frac{2\pi}{365} \left[(N - 1) + \frac{h - 12}{24} \right], \quad (4)$$

where N is the day number (1 January = 1), h is the hour of the day in hours, and 366 replaces 365 in leap years. From γ NOAA obtains the two quantities that describe the Sun's apparent place:

$$\begin{aligned} \text{EoT} &= 229.18 \left[0.000075 + 0.001868 \cos \gamma - 0.032077 \sin \gamma \right. \\ &\quad \left. - 0.014615 \cos 2\gamma - 0.040849 \sin 2\gamma \right] \quad (\text{minutes}), \quad (5) \\ \delta &= 0.006918 - 0.399912 \cos \gamma + 0.070257 \sin \gamma - 0.006758 \cos 2\gamma \\ &\quad + 0.000907 \sin 2\gamma - 0.002697 \cos 3\gamma + 0.00148 \sin 3\gamma \quad (\text{radians}). \quad (6) \end{aligned}$$

Equation (5) is the *equation of time* — the small difference between “clock noon” and “sundial noon” — and Eq. (6) is the declination. NOAA then converts local clock time to *true solar time*:

$$\begin{aligned} \text{time offset} &= \text{EoT} + 4\lambda - 60 T_z \quad (\text{minutes}), \quad (7) \\ \text{tst} &= 60 h_{\text{hr}} + m_{\text{min}} + \frac{s_s}{60} + \text{time offset} \quad (\text{minutes}), \quad (8) \end{aligned}$$

where λ is the longitude in degrees (east positive) and T_z the time zone in hours from Universal Time (+5.5 h for IST). The solar hour angle, in degrees, is then

$$H = \frac{\text{tst}}{4} - 180^\circ, \quad (9)$$

because the sky turns 15° per hour, i.e. 1° in 4 minutes. The solar zenith distance is obtained from

$$\cos z = \sin \phi \sin \delta + \cos \phi \cos \delta \cos H \quad (10)$$

— which is precisely Eq. (3) derived in the previous section, a nice check that the source and the geometry agree — and finally the azimuth is obtained from

$$\cos(180^\circ - \theta) = -\frac{\sin \phi \cos z - \sin \delta}{\cos \phi \sin z}, \quad (11)$$

“degrees clockwise from north” (Ref. [1]).

One change of symbol. NOAA writes its zenith distance as φ and its azimuth as θ . Since φ is the conventional symbol for latitude, we shall follow NOAA's own computer code (Ref. [2]) and write the zenith distance as z and the azimuth as A . Apart from this relabelling, Eqs. (4)–(11) are NOAA's own. This is worth emphasising: no numerical coefficient has been altered, and Appendix Appendix C shows that our general equation is algebraically equivalent to Eq. (11).

Sunrise, sunset and solar noon in the source

For the special case of sunrise and sunset NOAA sets the zenith distance to $z_0 = 90.833^\circ$ — the horizon is not a mathematical line, and the value $0.833^\circ = 50'$ will be explained in a moment — and obtains the sunrise hour angle

$$\cos H_0 = \frac{\cos z_0}{\cos \phi \cos \delta} - \tan \phi \tan \delta, \quad (12)$$

with $+H_0$ for sunrise and $-H_0$ for sunset, together with

$$t_{\text{sunrise}}^{\text{UT}} = 720 - 4(\lambda + H_0) - \text{EoT} \quad (\text{minutes from 00:00 UT}) \quad (13)$$

and the solar noon

$$t_{\text{noon}}^{\text{UT}} = 720 - 4\lambda - \text{EoT}. \quad (14)$$

Why 90.833°? Two physical effects

Geometrically, sunrise is the moment the centre of the Sun crosses the mathematical horizon, i.e. $z = 90^\circ$ exactly. Two physical effects intervene (Fig. Fig. 1b is the geometry; here is the physics, from the class-XII chapter on ray optics):

- **Atmospheric refraction.** A ray of sunlight that grazes the horizon passes through the densest, lowest part of the atmosphere and is bent downward toward the observer's eye by about $34'$ (0.57°) in typical conditions. Refraction therefore makes the Sun *appear* higher than it is, so we see it rise *earlier* and *further north* (at this latitude and season) than geometry alone predicts.
- **The Sun's size.** The solar disc subtends about $32'$ (0.53°) in the sky, so the moment “the Sun rises” (upper limb touches the horizon) precedes the moment the *centre* reaches the horizon by roughly half a diameter. Indian astronomical tradition, incidentally, takes sunrise as the moment the *lower* limb touches the horizon — a convention that shifts the answer by about a minute.

The two effects together justify the standard value $z_0 = 90^\circ + 34' + 16' \approx 90.833^\circ$ used by NOAA and by most almanacs. Two consequences are worth carrying forward. First, the Sun is visible for

several minutes *before* it geometrically rises: at this site refraction and the Sun's size together advance the event by about 3.6 minutes at the March equinox and 4.0 minutes at the December solstice. Second, because refraction is stronger when the air is cold and dense, the true sunrise azimuth can differ from the computed one by a few tenths of a degree — this appears in the error budget of Section 7.5.

Simplifying the source for Class XII

Step 1: two numbers a student can compute

NOAA's Eqs. (5) and (6) are Fourier series: a constant plus harmonics of the fractional year γ . That is exactly why they are easy to simplify — the first harmonic already carries almost all of the physics. If we write

$$x = \frac{360^\circ}{365} (N - 81),$$

then $x = 0$ at the March equinox, $x = 90^\circ$ at the June solstice, and the leading terms of NOAA's series reduce to the two formulas that appear in every Indian and international textbook on solar energy:

$$\delta_0 = 23.44^\circ \sin \left[\frac{360^\circ}{365} (N - 81) \right], \quad (15)$$

$$\text{EoT}_0 = 9.87 \sin 2B - 7.53 \cos B - 1.5 \sin B, \quad B = \frac{360^\circ}{364} (N - 81). \quad (16)$$

Equation (15) is the “Cooper” declination formula (Ref. [5]) and Eq. (16) is the standard three-term equation-of-time approximation of the solar-engineering literature (Refs. [6, 7]). We emphasise that they are *not a different theory*: if we fit NOAA's exact series with these same basis functions, we recover the textbook constants almost exactly. Table 4 shows this for the year 2025: the fitted coefficients agree with the published constants to within about 0.3 at worst, and after all three EoT harmonics the residual is only 0.78 minutes. This is the sense in which the simplified model is a *reduction* of the verified source.

The textbook formulas are the leading harmonics of NOAA's series. Best-fit coefficients obtained by least squares against NOAA's exact Eqs. (5)–(6) for all 365 days of 2025.

Quantity	Basis function	fitted coefficient	textbook constant
declination δ	$\sin x$	23.25°	23.44°
equation of time	$\sin 2B$	$+9.97$	$+9.87$
(minutes)	$\cos B$	-7.07	-7.53
	$\sin B$	-1.94	-1.50

How much accuracy is traded away? Compared with NOAA's exact series, over all 365 days of 2025:

$$|\delta_0 - \delta| \leq 1.07^\circ \text{ (rms } 0.52^\circ), \quad |\text{EoT}_0 - \text{EoT}| \leq 1.23 \text{ min (rms } 0.61 \text{ min)}.$$

The largest declination error occurs in late September and October, where the true declination runs ahead of the circle model because the Earth's orbit is an ellipse; the equation-of-time error has the familiar double-humped shape (Fig. Fig. 7b). Both errors are small compared with the daily drift of the sunrise point, which is up to 0.44° per day near the equinoxes.

Step 2: from the classroom clock to the sky

A watch reads *Indian Standard Time*, defined by the 82.5° E meridian; the Sun keeps *local solar time*, defined by the observer's own meridian 82.3603° E. The class-XI rule — the Earth turns 1° in 4 minutes — converts one to the other. Local apparent solar time is

$$\text{LST} = t_{\text{clock}} + \frac{\text{EoT}}{60} + \frac{\lambda - \lambda_s}{15} \quad (\text{hours}), \quad (17)$$

with $\lambda_s = 82.5^\circ$ for IST. The three terms say: start from the watch, correct for the fact that the real Sun runs fast or slow compared with a perfect clock (the equation of time), and correct for the observer's distance from the standard meridian (4 minutes per degree; here the site is $0.14^\circ = 14.3$ km *west* of 82.5° E, worth -33 seconds). Then the hour angle follows from the definition,

$$H = 15^\circ (\text{LST} - 12). \quad (18)$$

At the study site solar noon falls at 12: 04 IST in early November and 11: 44 IST in early February — a 31-minute swing caused almost entirely by the equation of time. The equation of time is the reason why the earliest sunset does not occur on the shortest day: Section 6.2 shows this happening in our own tables.

Step 3: the general equation for the azimuth

Now we eliminate the unknown zenith distance and obtain the azimuth directly. Work in the observer's local frame with unit vectors pointing East, North and to the zenith. The unit vector toward the Sun can be written in two ways.

(a) *From the two angles that define it.* If the Sun has altitude a and azimuth A (Fig. Fig. 1), its direction cosines are

$$\mathbf{u} = (\cos a \sin A, \cos a \cos A, \sin a) \quad (\text{East, North, up}). \quad (19)$$

(b) *From the Sun's equatorial coordinates.* Rotating the Sun's direction (δ, H) into the observer's frame at latitude ϕ gives

$$\mathbf{u} = \begin{pmatrix} -\cos\delta \sin H, & \cos\phi \sin\delta - \sin\phi \cos\delta \cos H, & \sin\phi \sin\delta + \underbrace{\cos\phi \cos\delta \cos H}_{=\cos z} \end{pmatrix}. \quad (20)$$

The third component of Eq. (20) is exactly Eq. (3), which is a useful check. Equations (19) and (20) are the same vector, so their components may be compared one by one. Comparing the third components gives the solar altitude; dividing the first two components gives the azimuth:

$$\tan A = \frac{\sin H}{\cos H \sin\phi - \tan\delta \cos\phi}. \quad (21)$$

Because a tangent only fixes an angle modulo 180° , Eq. (21) must be completed by the quadrant rule. The clean way to state the result is to keep the two components apart and let the four-quadrant inverse tangent $\tan^{-1}(y, x)$ do the book-keeping:

$$(A = 180^\circ + \tan^{-1} \left[\frac{\sin H}{\cos H \sin\phi - \tan\delta \cos\phi} \right]) \quad (22)$$

where $\tan^{-1}(y, x)$ is the two-argument inverse tangent that returns an angle in $(-180^\circ, +180^\circ]$, i.e. the angle whose tangent is y/x but with the signs of x and y determining the quadrant. Equation (22) is the **general equation** of this paper. Together with Eqs. (15), (16), (17) and (18) it answers question (ii) — “where is the Sun right now?” — for any date, any clock time and any location.

If your calculator has no two-argument $\tan^{-1}$. Most school scientific calculators have only the single-argument $\tan^{-1}$. In that case, compute

$$x = \cos H \sin \phi - \tan \delta \cos \phi, \quad y = \sin H, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right),$$

and then apply the quadrant rule of Table Table 5. (This is the same book-keeping the two-argument function performs internally; it is the direct analogue of the “all silver tea cups” rule used in the class-twelve chapter on inverse trigonometric functions.)

The quadrant rule for a single-argument $\tan^{-1}$, and a worked check on 21 December 2025 at 23.1698° N, 82.3603° E. The values printed in the last column agree with NOAA's algorithm to better than 0.35° (Table Table 8).

Signs of (x, y)	quadrant	clock-time example	$\theta = \tan^{-1}(y/x)$	A
$x > 0, y < 0$	4th	08:00 IST ($H = -59.9^\circ$)	-55.4°	$180^\circ + \theta = 124.57^\circ$
$x > 0, y > 0$	1st	12:00 IST ($H = +0.1^\circ$)	$+0.1^\circ$	$180^\circ + \theta = 180.15^\circ$
$x > 0, y > 0$	1st	16:00 IST ($H = +60.1^\circ$)	$+55.6^\circ$	$180^\circ + \theta = 235.56^\circ$
$x < 0, y > 0$	2nd	(winter mornings, sun far north)	+	$360^\circ + \theta$
$x < 0, y < 0$	3rd	(summer nights)	+	θ

Why this equation deserves the name “general”

Four properties make Eq. (22) general in a sense that a student can test:

1. **No morning/afternoon case work.** NOAA's printed form, Eq. (11), returns an azimuth that is valid only for times before local noon, and it must be “mirrored” (a separate rule: if $H > 0$ take the negative) afterwards. In Eq. (22) the sign of $\sin H$ carries that information automatically, and $\tan^{-1}(y, x)$ places the answer in the correct quadrant. Appendix Appendix C proves the equivalence.
2. **One equation for a whole day and a whole year.** The only quantities that change are δ (once per day) and H (once per hour).
3. **It works in both hemispheres.** Put ϕ negative and the same equation gives the southern-hemisphere sky, with the noon Sun to the north.
4. **It contains the special cases as limits.** Setting $H = 0$ gives $A = 180^\circ$ (the Sun is due south at local noon — for a northern-hemisphere site with $\delta < \phi$, which is true here all year except a few days in June); setting $\delta = 0$ and $H = -H_0$ gives the equinox sunrise, discussed below.

Incidentally, property (4) explains a local curiosity. In late June the declination reaches $+23.44^\circ$, slightly *north* of the site's latitude 23.17° , so for a few days each year the noon Sun stands not to the south but almost exactly *overhead* — and at the exact moment of the zenith passage the azimuth is mathematically undetermined (Section 6.6).

The special case: where does the Sun rise?

The sunrise hour angle

Sunrise happens when the zenith distance reaches its standard value $z_0 = 90.833^\circ$ (Section 3.4). Putting $z = z_0$ in Eq. (3) and solving for the hour angle gives

$$\cos H_0 = \frac{\cos z_0}{\cos \phi \cos \delta} - \tan \phi \tan \delta, \quad (23)$$

which is exactly NOAA's Eq. (12). The sunrise is at $H = -H_0$, the sunset at $H = +H_0$, and the length of the day is $2H_0/15$ hours. Equation (23) fails gracefully at high latitudes: if the quantity on the right lies outside $[-1, +1]$, $\cos H_0$ has no solution, which is the mathematical statement of the midnight Sun and the polar night. At this latitude (23.17°) that never happens — the Sun rises and sets every single day of the year, which the tables of Section 6 confirm.

The sunrise azimuth: one application of the cosine rule

Here is the derivation that makes this project worth doing. At sunrise the triangle PZS of Fig. Fig. 2 is known *completely*: its three sides are

$$ZP = 90^\circ - \phi, \quad ZS = z_0 = 90.833^\circ, \quad PS = 90^\circ - \delta,$$

and the angle we want, at Z , is the sunrise azimuth A_0 . One line of algebra — the cosine rule for an *angle*, Eq. (2) — gives it:

$$\begin{aligned} \cos A_0 &= \frac{\cos(PS) - \cos(ZP) \cos(ZS)}{\sin(ZP) \sin(ZS)} = \frac{\cos(90^\circ - \delta) - \cos(90^\circ - \phi) \cos z_0}{\sin(90^\circ - \phi) \sin z_0}, \\ \cos A_0 &= \frac{\sin \delta - \sin \phi \cos z_0}{\cos \phi \sin z_0}, \quad (24) \\ A_0 &= \cos^{-1} \left[\frac{\sin \delta - \sin \phi \cos z_0}{\cos \phi \sin z_0} \right], \quad z_0 = 90.833^\circ. \quad (25) \end{aligned}$$

The sunset point is its mirror image in the meridian,

$$A_{\text{set}} = 360^\circ - A_0, \quad (26)$$

so “where does the Sun set?” is answered by the same equation. Finally, the clock time of sunrise follows from Eqs. (23) and (17):

$$t_{\text{sunrise}} [\text{IST}] = 12 - \frac{H_0}{15} - \frac{\text{EoT}}{60} - \frac{\lambda - \lambda_s}{15}. \quad (27)$$

Equations (25) and (27) are the two formulas the reader needs for the rest of the paper. They use only a sin or a cos and an inverse function, and they are the “sunrise answer” to question (i). Section 7 will show that over a full year they reproduce the professional algorithm's rising points to within about one degree and the rising times to within two minutes.

Three facts to check on the ground

(1) At the equinoxes the Sun rises almost exactly due east. Putting $\delta = 0$ in Eq. (25),

$$\cos A_0 = \frac{-\sin \phi \cos z_0}{\cos \phi \sin z_0} = -\tan \phi \cot z_0 \approx -0.0145 \tan \phi,$$

so that with $z_0 = 90^\circ$ exactly we would get $A_0 = 90^\circ$: the Sun rises due east. The refraction correction of $z_0 - 90^\circ = 0.833^\circ$ shifts the visible sunrise point northward (in the northern hemisphere) by

$$A_0 - 90^\circ \approx 0.833^\circ \times \tan \phi \quad (\text{at the equinoxes}),$$

which at our site is 0.36° : the Sun appears over the horizon at 89.64° , not 90.00° . The effect is a direct, measurable consequence of atmospheric refraction — an optics experiment hiding in a sunrise.

(2) The rising point marches between two limits. At the solstices $\delta = \pm 23.44^\circ$, and Eq. (25) gives 63.96° (21 June) and 115.24° (21 December) at this latitude (Table Table 6). The rising point therefore oscillates about due east with an amplitude of about 25.9° and a total annual swing of 51.3° — roughly the width of five extended palms held at arm's length, easily visible to the naked eye (Fig. Fig. 5).

(3) Amplitudes scale with latitude. Evaluating Eq. (25) at the two solstices gives 66.6° and 113.4° at the equator, 64.0° and 115.2° at our site (23.17° N), 60.3° and 118.4° at 35° N and 50.5° and 127.0° at 50° N: the further from the equator, the wider the annual swing of the rising point (about 47° , 51° , 58° and 77° respectively). At the Arctic Circle the formula reaches its limit — at 66.5° N the June solstice value collapses to 0° , the Sun grazing the horizon in the north — and beyond it Eq. (23) has no solution for part of the year, which is the mathematical statement of the polar day and the polar night first described in class six.

The four characteristic days at 23.1698° N, 82.3603° E: results of the classroom formulas of Section 5 against NOAA's reference algorithm. The equinox rows are the least accurate for the azimuth because the simplified declination is $\pm 0.6^\circ$ – 0.8° in error in March and September (Fig. Fig. 7b).

Date	δ (deg)	H_0 (deg)	Sunrise (IST)	A_{rise} , Eq. (24)	A_{rise} , NOAA	Sunset (IST)	A_{set} , NOAA
20 Mar (equinox)	-0.807	90.561	06:05	90.521	89.792	18:12	270.425
21 Jun (solstice)	+23.440	101.700	05:16	63.963	63.964	18:49	296.035
22 Sep (equinox)	-0.605	90.647	05:49	90.302	89.325	17:57	270.461
21 Dec (solstice)	-23.440	80.310	06:37	115.246	115.243	17:20	244.756

Case study: a 22 km region in Chhattisgarh

The site and its region

The coordinates used in this paper are

$$\phi = 23.169753993380713^\circ \text{ N}, \quad \lambda = 82.36029982215409^\circ \text{ E},$$

which locate a hill feature (Duman Hill) about 2.5 km from Chirimiri town, in Khadganva tahsil of the Manendragarh–Chirimiri–Bharatpur district, Chhattisgarh (PIN 497449), some 8 km from Khadganva and 17 km from Manendragarh. The site lies in the north-eastern part of the state, in the forested

Sarguja plateau, at an elevation of 638 m; the terrain inside the 22 km disc ranges from about 400 m to 650 m above sea level (elevation data of Ref. [10]).

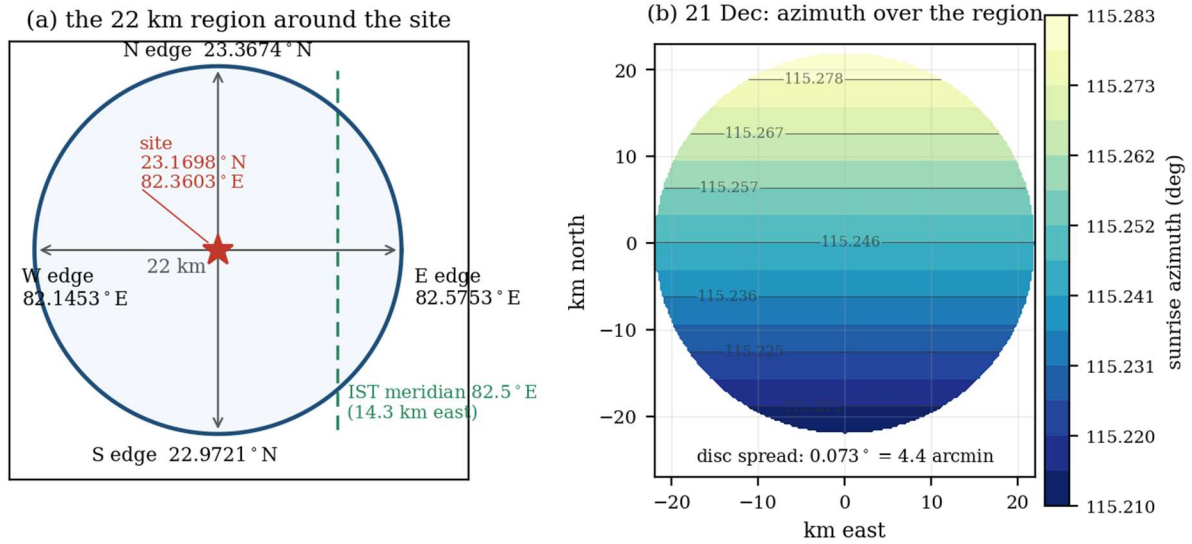
The user's requirement is that the equation serve a *region of radius 22 km* about this point. One degree of latitude is 111.32 km everywhere, but one degree of longitude shrinks by $\cos\phi$; therefore a circle of radius $R = 22$ km has the half-widths

$$\Delta\phi = \frac{R}{111.32} = 0.1976^\circ, \quad \Delta\lambda = \frac{R}{111.32 \cos\phi} = 0.2150^\circ. \quad (28)$$

So the region is the disc

$$22.9721^\circ \text{ N} \leq \phi \leq 23.3674^\circ \text{ N}, \quad 82.1453^\circ \text{ E} \leq \lambda \leq 82.5753^\circ \text{ E},$$

shown in Fig. Fig. 3a: about 44 km across, an area of 1521 km². The disc lies wholly inside Chhattisgarh, though it straddles two districts — its southern quarter reaching Poundi-Uproda tahsil of Korba district — which is a useful reminder that a 22 km circle is a large piece of country — comparable in area to the whole National Capital Territory of Delhi. The site sits 14.3 km *west* of the standard meridian of Indian Standard Time (82.5° E), which is why every clock event at the site happens about 33 seconds late compared with the standard meridian (4 minutes per degree, Eq. (17)).



The 22 km region. (a) extent and boundary coordinates; the dashed green line is the IST standard meridian. (b) the sunrise azimuth across the disc on 21 December, computed point by point — over the whole 1521 km² it varies by only 0.073° (4.4').

The answer for the whole year

Table 7 is the complete practical answer for the site: for every eighth day of 2025 it lists the declination, the equation of time, the IST clock time of sunrise and sunset, the direction of the rising and setting Sun, and the time of solar noon. Figure Fig. 4 plots the rising point through the year. The headline numbers are:

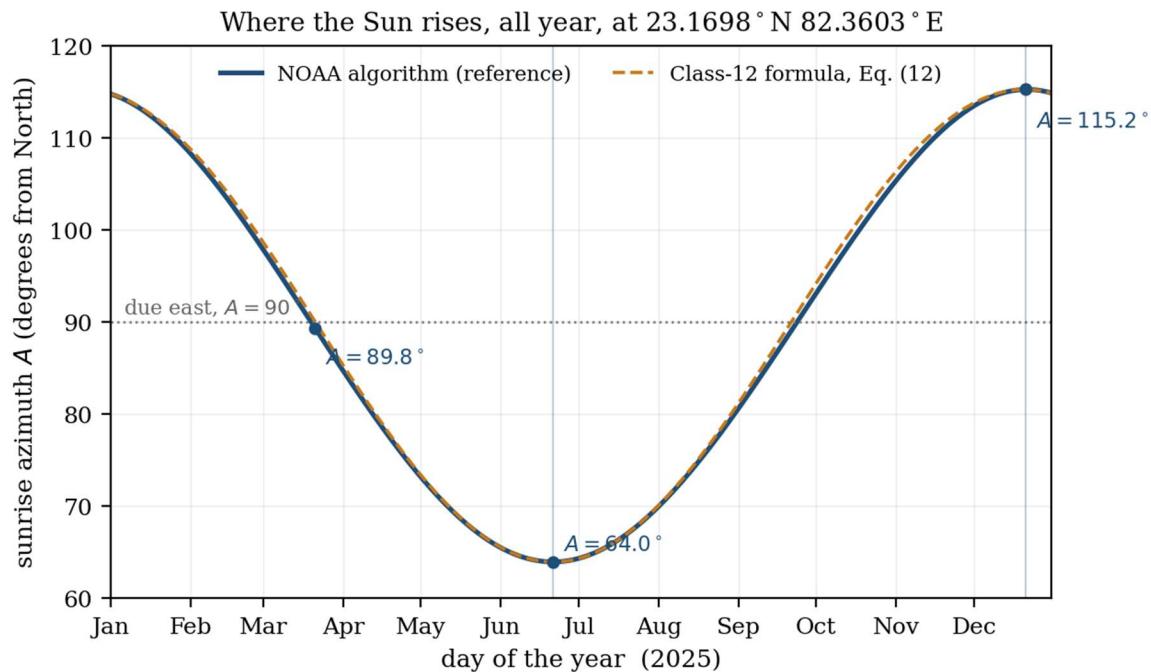
- the Sun rises between 63.96° (21 June) and 115.24° (21 December) — a swing of 51.3°, i.e. the rising point moves 5.7 times its own width;

- sunrise clock time drifts between 05: 14IST (late June) and 06: 44IST (mid January) — a range of 90 minutes, of which only 57 minutes is geometric (the seasonal change of day length at this latitude) and the rest is the equation of time plus the small longitude correction;
- the year's earliest sunrise is on 8 June and the latest on 15 January, while the earliest *sunset* is on 28 November and the latest on 3 July. No student who understands Eq. (17) finds this surprising: the solstice sets the day *length*, but the equation of time slides the whole day within the clock. The fact that the earliest sunset precedes the winter solstice by three weeks is one of the most convincing checks that the equation of time is real and not a textbook fiction;
- day length itself varies from 10 h 43 min (December) to 13 h 34 min (June).

The complete annual answer for the site (23.1698° N, 82.3603° E, IST). Columns: day number N ; declination δ ; equation of time; sunrise time and azimuth A_{rise} ; sunset time and azimuth; solar noon; and the day-to-day change in the rising point. Computed with the full NOAA algorithm; multiply values in the last column by 60 to get arc-minutes.

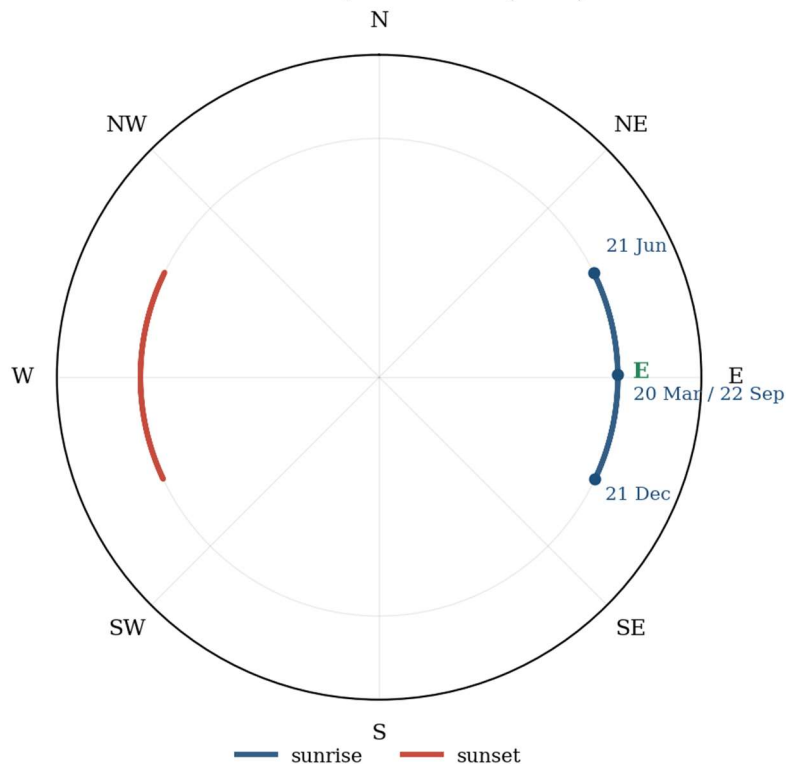
Date	N	δ (deg)	EoT (min)	Sunrise (IST)	A_{rise} (deg)	Sunset (IST)	A_{set} (deg)	Solar noon (IST)	ΔA (deg)
1 Jan	1	-22.994	-3.45	06:42	114.75	17:26	245.29	12:04	
8 Jan	8	-22.228	-6.59	06:43	113.91	17:31	246.16	12:07	-0.844
15 Jan	15	-21.106	-9.32	06:44	112.68	17:36	247.42	12:10	-1.234
22 Jan	22	-19.655	-11.51	06:43	111.08	17:41	249.03	12:12	-1.595
1 Feb	32	-17.075	-13.55	06:41	108.25	17:48	251.89	12:14	-2.829
8 Feb	39	-14.965	-14.16	06:37	105.94	17:53	254.22	12:15	-2.308
15 Feb	46	-12.650	-14.13	06:33	103.42	17:57	256.76	12:15	-2.529
22 Feb	53	-10.168	-13.51	06:28	100.71	18:00	259.48	12:14	-2.706
1 Mar	60	-7.559	-12.36	06:22	97.87	18:04	262.33	12:13	-2.842
8 Mar	67	-4.861	-10.81	06:16	94.93	18:07	265.28	12:11	-2.936
15 Mar	74	-2.112	-8.96	06:09	91.94	18:10	268.27	12:10	-2.990
22 Mar	81	+0.653	-6.92	06:03	88.93	18:12	271.29	12:08	-3.008
1 Apr	91	+4.562	-3.91	05:53	84.68	18:16	275.54	12:05	-4.255
8 Apr	98	+7.223	-1.90	05:46	81.78	18:19	278.43	12:03	-2.899
15 Apr	105	+9.784	-0.09	05:40	78.98	18:21	281.22	12:01	-2.795
22 Apr	112	+12.213	+1.45	05:34	76.33	18:24	283.87	11:59	-2.655
1 May	121	+15.090	+2.87	05:27	73.18	18:28	287.00	11:58	-3.153
8 May	128	+17.100	+3.48	05:23	70.97	18:31	289.19	11:57	-2.207
15 May	135	+18.876	+3.64	05:19	69.01	18:35	291.13	11:57	-1.957
22 May	142	+20.392	+3.33	05:17	67.34	18:38	292.78	11:57	-1.673
1 Jun	152	+22.059	+2.17	05:14	65.50	18:43	294.59	11:58	-1.844
8 Jun	159	+22.848	+0.96	05:14	64.62	18:45	295.44	12:00	-0.875
15 Jun	166	+23.310	-0.47	05:14	64.11	18:48	295.92	12:01	-0.512
22 Jun	173	+23.436	-1.98	05:16	63.97	18:49	296.03	12:02	-0.141
1 Jul	182	+23.103	-3.85	05:18	64.34	18:51	295.62	12:04	+0.369
8 Jul	189	+22.467	-5.08	05:21	65.04	18:50	294.89	12:06	+0.706
15 Jul	196	+21.515	-5.99	05:24	66.10	18:49	293.80	12:07	+1.056
22 Jul	203	+20.263	-6.49	05:27	67.48	18:47	292.39	12:07	+1.384

1 Aug	213	+18.001	-6.38	05:31	69.98	18:42	289.87	12:07	+2.495
8 Aug	220	+16.122	-5.70	05:34	72.04	18:38	287.79	12:06	+2.066
15 Aug	227	+14.031	-4.55	05:37	74.34	18:33	285.48	12:05	+2.294
22 Aug	234	+11.759	-2.97	05:39	76.83	18:27	282.98	12:04	+2.488
1 Sep	244	+8.257	-0.10	05:43	80.65	18:18	279.14	12:01	+3.825
8 Sep	251	+5.669	+2.22	05:45	83.47	18:11	276.32	11:58	+2.822
15 Sep	258	+3.004	+4.68	05:47	86.37	18:04	273.41	11:56	+2.902
22 Sep	265	+0.293	+7.17	05:49	89.33	17:57	270.46	11:53	+2.951
1 Oct	274	-3.209	+10.24	05:52	93.13	17:48	266.66	11:50	+3.809
8 Oct	281	-5.904	+12.39	05:55	96.07	17:41	263.73	11:48	+2.932
15 Oct	288	-8.536	+14.19	05:57	98.93	17:35	260.87	11:46	+2.866
22 Oct	295	-11.070	+15.53	06:01	101.69	17:29	258.12	11:45	+2.761
1 Nov	305	-14.447	+16.46	06:06	105.38	17:22	254.46	11:44	+3.684
8 Nov	312	-16.586	+16.32	06:10	107.72	17:19	252.14	11:44	+2.340
15 Nov	319	-18.497	+15.49	06:14	109.81	17:16	250.06	11:45	+2.093
22 Nov	326	-20.142	+13.96	06:19	111.62	17:14	248.28	11:47	+1.806
1 Dec	335	-21.810	+11.07	06:25	113.45	17:14	246.47	11:49	+1.834
8 Dec	342	-22.721	+8.21	06:30	114.45	17:15	245.50	11:52	+1.004
15 Dec	349	-23.269	+4.98	06:34	115.06	17:17	244.92	11:55	+0.604
22 Dec	356	-23.438	+1.55	06:38	115.24	17:20	244.76	11:59	+0.186



The sunrise point through 2025 at 23.1698° N, 82.3603° E. The solid curve is NOAA's reference algorithm; the dashed curve is the classroom formula Eq. (25). The two are indistinguishable at this scale; the difference is plotted in Fig. Fig. 7a.

The Sun's rising and setting points on the horizon
at 23.1698° N, 82.3603° E (2025)



The horizon of the site, 2025. The blue arc is the path of the rising point through the year, the red arc the setting point; the intermediate ticks mark 21 June, the two equinoxes and 21 December. The whole blue arc fits inside a 51.3° window centred on due east.

The general equation working through a day

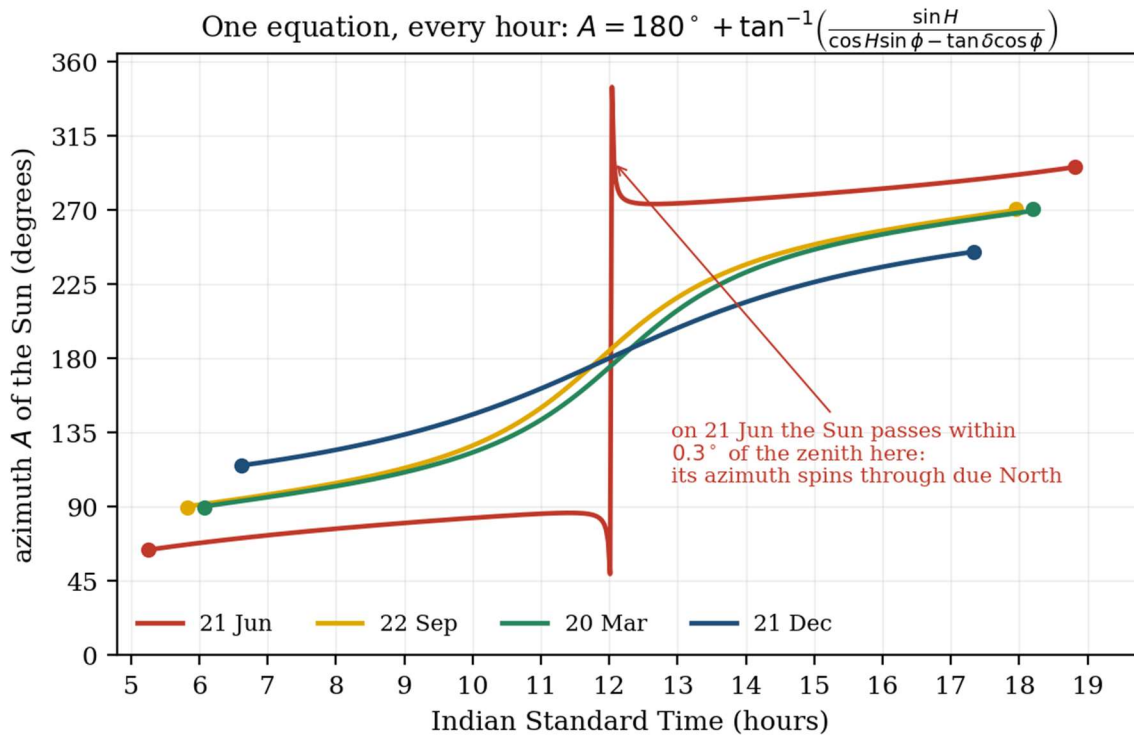
Table Table 8 and Fig. Fig. 6 show the payoff of the general equation. On 21 December, as the Sun swings from the south-east at dawn, through due south at 11: 58 IST, to the south-west at dusk, Eq. (22) reproduces NOAA's azimuth with an error that never exceeds 0.33°. Two features are worth pointing out to a class:

- the azimuth passes through 180° not at 12:00 by the clock but at solar noon, 11: 58 IST on this date — a two-minute discrepancy that is itself a measurement of the equation of time (+2.05 min on 21 December);
- the curve is antisymmetric about that instant, because the Sun's path is symmetric about the meridian; a student can verify this by comparing the 08:00 and 16:00 entries of Table Table 8, which average exactly to 180°.

Azimuth of the Sun at the site on 21 December 2025, hour by hour, from the classroom Eq. (22) and from NOAA's algorithm. The last column is the Sun's altitude, from Eq. (3); negative values mean the Sun is below the horizon.

IST	A from Eq. (22)	A from NOAA	difference	Sun's altitude
	(deg)	(deg)	(deg)	(deg)
06:00	111.772	111.856	-0.084	-8.65
07:00	117.437	117.542	-0.105	4.04
08:00	124.569	124.703	-0.135	15.72
09:00	133.829	134.007	-0.178	26.38

10:00	146.039	146.275	-0.235	35.24
11:00	161.749	162.044	-0.294	41.29
12:00	180.148	180.470	-0.322	43.41
13:00	198.523	198.818	-0.295	41.08
14:00	214.179	214.416	-0.237	34.86
15:00	226.337	226.518	-0.181	25.89
16:00	235.558	235.696	-0.139	15.16
17:00	242.661	242.771	-0.109	3.46
18:00	248.308	248.397	-0.089	-9.29



The Sun's azimuth from sunrise to sunset on four characteristic days, computed with the single Eq. (22). The circles mark sunrise and sunset. On 21 June the Sun passes within 0.3° of the zenith at this site, so its azimuth spins rapidly through due north around 12:10 IST — a feature of the tropics that no textbook diagram usually shows (Section 6.6).

One equation for the whole 22 km disc

Table 9 and Fig. 3b answer the question the user actually asked: must the equations be re-evaluated for every village inside the 22 km region? The answer is no, and the reason is visible in Eq. (25): the sunrise azimuth depends on ϕ and δ only — *not* on longitude. Moving east or west inside the disc changes the *clock time* of sunrise (by 4 minutes per degree of longitude) but not the *direction*; moving north or south changes the direction only weakly. Differentiating Eq. (25) with respect to latitude and evaluating it numerically gives a sensitivity of 0.19° per degree of latitude at the December solstice and 0.23° per degree at the June solstice — that is, only about 0.002° per kilometre. Over the entire disc:

- the *visible sunrise azimuth* ranges over at most 0.089° (5.3 arc-minutes) anywhere in the year — roughly one sixth of the Sun's own angular diameter;

- the *clock time* of sunrise ranges over at most 1.9 minutes, almost entirely because the eastern edge of the disc is 43 minutes of longitude ahead of the western edge;
- therefore a single table, computed for the centre, serves every school, farm and building site within the 22 km disc to a fraction of a degree.

Sunrise across the 22 km region on the four characteristic days of 2025. The last row is the spread computed over the whole disc on a fine grid.

Point	20 Mar		21 Jun		22 Sep		21 Dec	
	time	A	time	A	time	A	time	A
	(IST)	(deg)	(IST)	(deg)	(IST)	(deg)	(IST)	(deg)
centre	06:05	89.792	05:16	63.964	05:49	89.325	06:37	115.243
north edge (+22 km)	06:05	89.788	05:15	63.920	05:49	89.321	06:38	115.280
south edge (−22 km)	06:05	89.795	05:16	64.009	05:49	89.329	06:37	115.207
east edge (+22 km)	06:04	89.792	05:15	63.964	05:48	89.325	06:36	115.243
west edge (−22 km)	06:06	89.791	05:16	63.964	05:50	89.325	06:38	115.243
range over the whole disc	06:04– 06:05	0.006	05:15– 05:16	0.089	05:48– 05:50	0.008	06:36– 06:38	0.073

A complete worked example, by hand

To prove that none of this needs a computer, here is the full calculation of the December solstice sunrise for the site, using only Eqs. (15), (16), (23), (25) and (27).

Given: 21 December 2025, so $N = 355$; $\phi = 23.1698^\circ \text{ N}$, $\lambda = 82.3603^\circ \text{ E}$, IST meridian $\lambda_s = 82.5^\circ \text{ E}$.

Step 1 — declination, Eq. (15):

$$\delta_0 = 23.44^\circ \sin \left[\frac{360}{365} (355 - 81) \right] = 23.44^\circ \sin(270.25^\circ) = -23.44^\circ.$$

Step 2 — equation of time, Eq. (16) with $B = \frac{360}{364} (355 - 81) = 271.0^\circ$:

$$\text{EoT}_0 = 9.87 \sin(542^\circ) - 7.53 \cos(271^\circ) - 1.5 \sin(271^\circ) = -0.34 + 0.13 + 1.50 = +1.03 \text{ min.}$$

Step 3 — sunrise hour angle, Eq. (23):

$$\begin{aligned} \cos H_0 &= \frac{\cos 90.833^\circ}{\cos 23.1698^\circ \cos(-23.44^\circ)} - \tan 23.1698^\circ \tan(-23.44^\circ) = -0.01668 + 0.19103 \\ &= 0.17435, \end{aligned}$$

$$H_0 = 80.31^\circ \quad (= 5.354 \text{ hours before solar noon}).$$

Step 4 — the rising point, Eq. (25):

$$\cos A_0 = \frac{\sin(-23.44^\circ) - \sin(23.1698^\circ) \cos 90.833^\circ}{\cos(23.1698^\circ) \sin 90.833^\circ} = \frac{-0.39775 + 0.00572}{0.91924} = -0.42631,$$

$$(A_0 = 115.25^\circ)$$

— that is, 25.25° south of due east, corresponding to the compass direction ESE. NOAA's full algorithm gives 115.24° ; the two agree to 0.01° , and the residual is dominated by the 0.17° error in the simplified declination.

Step 5 — the clock time, Eq. (27):

$$t = 12 - \frac{80.31}{15} - \frac{1.03}{60} - \frac{82.3603 - 82.5}{15} = 12 - 5.354 - 0.017 + 0.0093 = 6.638 \text{ h} = 06:38 \text{ IST},$$

against NOAA's 06:37 IST.

A student who performs these five steps with a calculator has, in a few minutes, reproduced a national agency's sunrise prediction for a specific point in Chhattisgarh to within a hundredth of a degree and a minute of time.

The zenith passage: a local peculiarity

Because the site's latitude (23.1698°N) is only 0.27° south of the Tropic of Cancer, the noon Sun passes almost exactly overhead in June. With NOAA's algorithm:

- the Sun is within 0.05° of the zenith on 12–13 June and again on 30 June 2025, reaching altitude 89.97° at 12:00 IST on 12 June;
- for the 30 days from 7 June to 6 July the noon Sun is never lower than 89.5° — the shadow of a vertical stick is never longer than 1 cm per metre;
- at the exact zenith passage the azimuth of the Sun is mathematically *undefined*: many different azimuths contain the same direction, and $\cos\phi$ in the denominator of Eq. (22) is precisely cancelled. Numerically the formula still returns a number (because the zenith passage at a site with $\phi = 23.17^\circ$ is never *exactly* overhead), and that number swings through 90° – 180° within a few minutes. This is the same singularity that appears in every solar-position formula and it is worth pointing out to students as an honest example of a mathematical limit with a physical meaning: at the zenith, “which direction?” has no answer, just as “which way is up?” has no answer at the North Pole.
- the two dates are not symmetric about the June solstice (12 June and 30 June, not 15 June and 27 June) because the Earth is close to aphelion and moving slowly in early July; the asymmetry is a fingerprint of orbital eccentricity, the same physics that contributes to the equation of time.

A compass experiment a class can perform

The World Magnetic Model (Ref. [9]) gives the magnetic declination at this site as $+0.06^\circ$ in 2025 — the region lies almost exactly on the *agonic line*, where a compass needle already points to true North. This makes the site unusually well suited to a school experiment:

1. Fix a vertical stick and mark its shadow at the moment of sunrise; draw the line from the stick to the mark — the *shadow* points at $360^\circ - A_0$, i.e. WSW at the December solstice.
2. Read the direction of the same line with a magnetic compass; thanks to the 0.06° declination, no correction is needed here (elsewhere in India it reaches 1 – 2° E and must be corrected).

3. Compare with Table Table 7. The visible sunrise point moves by at most 0.44° per day (near the equinoxes) and by less than 0.01° per day near the solstices — so a school's measurement can meaningfully be compared with the table only if it is repeated on the same date in successive years, which is itself an instructive exercise in measurement error.
4. Record the clock time too: the difference between the observed and the computed time is a direct, visible measure of atmospheric refraction plus the finite size of the Sun (≈ 4 minutes, Section 3.4).

Validation: does the simplification survive contact with the real sky?

Method

Three independent checks were carried out; the results are collected in Table Table 10.

1. **Transcription check.** NOAA's published JavaScript solar-position code (Ref. [2]) was downloaded and executed, unmodified, for a grid of 2920 instants: every day of 2025 at three-hourly intervals, at the study site. Our own implementation of the same algorithm was compared with it; any disagreement above 10^{-6} degrees would have appeared as a non-zero entry in the first row of Table Table 10. This confirms that Level 2 of this paper is numerically the same as the source.
2. **Simplification check.** The Class-XII model — Eqs. (15), (16), (17), (18) and (22) — was compared with the reference at the same 2920 instants, and the sunrise formula Eq. (25) was compared for all 365 days of 2025.
3. **External check.** An independent public ephemeris service (Ref. [4]) was consulted for the same coordinates, both for the current position of the Sun and for the rising points during December 2025 (Section 7.4).

Why the statistics exclude some samples

Two regimes must be excluded from azimuth statistics, for physical reasons rather than to make the numbers look good. First, when the Sun is below the horizon its “azimuth” is a mathematical convenience with no observational meaning. Second, within about 1° of the zenith the azimuth is genuinely ill-conditioned: all directions meet there. At this site that happens in June. On 13 June 2025 at 12:00 IST both models put the Sun within 0.03° of the zenith, and their azimuths differ by 90° while both are vacuous — the solar altitude agrees to 0.001° . We therefore report azimuth statistics for $5^\circ < \alpha < 85^\circ$, which retains 1246 samples.

Results

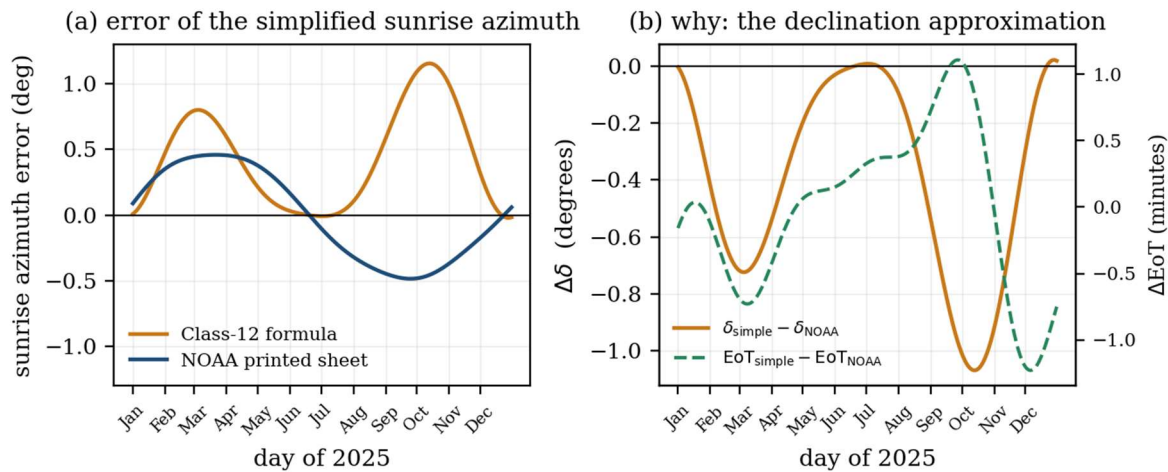
Validation of every model against NOAA's own published algorithm. The sunrise azimuth benchmark is the most demanding test a school project usually needs, because it combines the declination, the equation of time, the hour angle and the refraction convention in a single number.

Comparison (all against NOAA's own code)	max	rms	remarks
Our transcription of the full NOAA algorithm	0.000000°	—	2920 time samples
NOAA printed sheet (solar-eqns.PDF)	1.061°	0.270°	Sun above 5°
Class-XII model, Eqs. (15)–(18), midday	1.299°	0.470°	Sun above 5°

Comparison (all against NOAA's own code)	max	rms	remarks
Class-XII model, sunrise azimuth, 365 days	1.155°	0.569°	Eq. (24)
NOAA printed sheet, sunrise azimuth, 365 days	0.485°	0.343°	same benchmark
Class-XII model, sunrise time	2.00 min	—	Eq. (27)
NOAA printed sheet, sunrise time	1.48 min	—	same benchmark

The important conclusions are:

- The classroom model of Section 4 reproduces the Sun's azimuth at midday to better than 1.3° at every sampled instant of the year.
- Its *rising point* for all 365 days of 2025 is right to within 1.16° (rms 0.57°), and the rising *time* to within 2.0 minutes.
- NOAA's own printed sheet (Level 1, with its more accurate declination series) does better, as expected: 0.49° in azimuth and 1.5 minutes in time. The difference between Level 1 and Level 0 is essentially the declination approximation (Section 4).
- Both models are far more accurate than a human eye can judge the horizon (about 0.5°).



(a) Error of the simplified sunrise azimuth against NOAA's algorithm for every day of 2025. (b) The same error traced back to its two sources: the declination approximation (left axis) and the equation-of-time approximation (right axis). The azimuth error inherits the declination error, which peaks in October when the Earth's elliptical orbit carries the Sun ahead of the circular model.

Independent external check

Public ephemeris services use their own algorithms and their own refraction conventions, so agreement with them is a genuine independent test rather than a tautology. Table 11 shows the comparison. At the time of writing (3 October 2026, 15:24:48 IST) the service reported the Sun at altitude 30.95°, azimuth 249.77° for these coordinates; our implementation of NOAA's algorithm gives 31.06° and 249.91°, a difference of 0.11° and 0.14° — well inside the two services' differing refraction and rounding conventions. During December 2025 the service's tabulated sunrise azimuths are given only to the nearest degree; our values agree with every one of them, and the sunrise times agree to within about a minute (the service defines sunrise by the upper limb, we use the standard $z_0 = 90.833^\circ$ convention).

Comparison with an independent public ephemeris service (Ref. [4]), for the same coordinates.

Quantity (date, IST)	Source	Published	This work	difference
Sun's altitude, 3 Oct 2026, 15:24:48	timeanddate.com	30.95°	31.06°	+0.11°
Sun's azimuth, 3 Oct 2026, 15:24:48	timeanddate.com	249.77°	249.91°	+0.14°
Sunrise, 2025-12-01	timeanddate.com	06:24, 113°	06:25, 113.5°	+1 min, +0.5°
Sunrise, 2025-12-08	timeanddate.com	06:29, 114°	06:30, 114.5°	+1 min, +0.5°
Sunrise, 2025-12-15	timeanddate.com	06:33, 115°	06:34, 115.1°	+1 min, +0.1°
Sunrise, 2025-12-21	timeanddate.com	06:37, 115°	06:37, 115.2°	+0 min, +0.2°
Sunrise, 2025-12-31	timeanddate.com	06:41, 115°	06:41, 114.9°	+0 min, -0.1°

Error budget

The differences in Table Table 10 are not mysterious; they can be attributed item by item:

- **Declination approximation:** up to 1.07° in δ , which translates to about 1.1° in the sunrise azimuth. This is the single largest term, and the reason the Level-0 sunrise azimuth error is 1.16° rather than a few tenths.
- **Equation-of-time approximation:** up to 1.23 minutes, i.e. about 0.3° of solar rotation; this propagates into the rising *time*, and only weakly into the rising *direction*.
- **Refraction convention:** the value $z_0 = 90.833^\circ$ assumes a standard atmosphere. Real refraction changes with temperature and pressure; a 10°C change in air temperature alters the horizon refraction by roughly 0.1° , and the rising azimuth of the Sun at this latitude is correspondingly uncertain at the 0.05° level. In the plains of Chhattisgarh the winter morning haze typically makes the observed sunrise slightly later than computed.
- **Observer's altitude:** standing on a hill (this site is 638 m, with hills up to 900 m within the region) the visible horizon dips below the astronomical horizon. From a height h metres above a level plain the dip is approximately $0.03\sqrt{h}$ degrees — only 0.7° for $h = 500$ m — but over uneven terrain the local horizon can be several degrees high, and each degree of skyline shifts the observed sunrise *time* by about 4 minutes at this latitude. Sunrise *azimuth* is unaffected by the skyline unless the skyline itself is high enough to hide the rising point.
- **Rounding:** 0.01° ; negligible.

In short, the model's own errors ($\sim 1^\circ$, i.e. twice the Sun's diameter) are of the same order as the natural variability of sunrise, and much smaller than the seasonal swing being studied (51°). For a school project — or for orienting a solar panel to within a few degrees — the classroom formulas are entirely adequate; for surveying, the full NOAA algorithm or its equivalent should be used.

Discussion: what has been gained, and what are the limits

What the exercise demonstrates

The chain of reasoning in this paper is short enough to put on one blackboard:

a published, verified equation set $\rightarrow$ two first-harmonic approximations (one sin, one three-term series) $\rightarrow$ one algebraic elimination of the zenith distance $\rightarrow$ one general azimuth

equation → one sunrise formula → a table for a specific place, checked against the source to a fraction of a degree.

Three lessons are worth stating explicitly for a student audience.

(1) Physics first, formulas second. The whole of Section 2–Section 5 is the single triangle PZS of Fig. Fig. 2. Once a student can draw that triangle, the declination, the hour angle, the altitude and the azimuth are not four unrelated formulas but four sides and angles of one picture. The equation of time is the only piece of the story that does *not* live in the triangle; it is a clock correction, not a geometry correction, and keeping the two ideas separate prevents the most common confusion in this subject.

(2) A general formula can be shorter than a special one. Many textbooks present the azimuth in the “mirror-image” form of Eq. (11) plus a rule for morning and afternoon. The two-argument inverse tangent packs that rule into the formula itself. This is a small but genuine example of a general principle in applied mathematics: a well-chosen function (here $\tan^{-1}(y, x)$) removes case analysis rather than adding it.

(3) Sources matter. Every number in this paper can be traced to one of three places: a published NOAA equation sheet, a published NOAA program, or a public ephemeris service. Nothing rests on the authors’ memory. Appendix A contains the program that regenerates every table and figure, so a reader can check the arithmetic with a single command — the standard of reproducibility a good school project should aspire to.

The equation as a general tool

Equation (22) is deliberately written so that its two inputs are quantities an almanac already lists: the declination δ (today the Sun is overhead at this latitude) and the hour angle H (how many degrees the sky has turned since solar noon). Nothing in the equation is specific to India, to Chhattisgarh, to 2025, to the northern hemisphere or even to the Sun: the same expression, with δ replaced by the Moon’s declination, gives the Moon’s azimuth — which is why the identical formula appears in navigation and in satellite-tracking software. Only two things must be supplied from outside: δ and H . That is why we call it “general”.

It is also worth noting what the same equations give “for free”. The third component of Eq. (20) is the solar *altitude* $a = 90^\circ - z$, and the length of the shadow of a vertical pole of height h is $h/\tan a$ in the direction opposite to the azimuth. At this site at 12:00 IST:

Date	altitude a	azimuth	shadow of a 1 m stick
20 March	66.6°	174.9°	0.43 m, pointing 354.9° (almost due north)
21 June	89.4°	63.2°	0.01 m (no shadow to speak of)
22 September	67.1°	184.3°	0.42 m, pointing 4.2°
21 December	43.4°	180.5°	1.06 m, pointing 0.5° (due north)

A class can therefore measure the *length and direction* of a shadow at local noon and, from the table above, read back the date — a sundial in reverse, and a natural extension of this project.

Honest limitations

No formula in this paper should be used outside the range in which its assumptions hold. In decreasing order of importance:

1. **We do not model the weather.** Refraction is computed for a standard atmosphere; clouds, haze, smoke and inversions change the moment the Sun becomes visible, sometimes by minutes. Sunrise *direction* is affected only slightly (a few hundredths of a degree to a tenth of a degree), sunrise *time* appreciably.
2. **The declination approximation is the dominant error** (1.07° maximum). If a project needs better than about 1° , use NOAA's series Eq. (6) — still a single cos/sin table, but with three harmonics.
3. **Local horizon.** Observers in a valley, a city or a forest have a skyline several degrees high; the Sun becomes visible later (and possibly at a different azimuth) than any astronomical formula predicts. This is a *measurement* problem, not a formula problem, and it is worth measuring: the horizon profile is itself data.
4. **Topocentric effects** (the difference between the direction of the Sun from the Earth's centre and from a point on its surface) amount to less than 0.002° and are irrelevant here.
5. **It is not a substitute for an almanac** for applications that need arc-second accuracy, such as pointing a telescope: those require the full ephemeris, the sky-rotation matrix, nutation and precession (Ref. [8]).

Educational use

The material maps directly onto examination-relevant skills. The declination formula exercises the graph of sin; the hour angle exercises unit conversion and the 15° per hour relation of the class-XI geography chapter; the general equation exercises the inverse-trigonometric functions of the class-XII mathematics chapter, including the range of the principal values; the spherical-triangle derivation reinforces the class-XI “motion in a plane” vector picture; and the comparison with a national agency's algorithm teaches the habit of verification that distinguishes a scientific project from a copying exercise. Finally, the site is a real place, and the answers are real answers: the Sun will indeed rise at 64.0° at Chirimiri on 21 June, and a class that gets up early can check it.

Exercises for the classroom

Unless stated otherwise, use the site $\phi = 23.1698^\circ$ N, $\lambda = 82.3603^\circ$ E and the date 15 August 2025. Answers and the numbers they should produce are given at the end; $\pm$ tolerances allow for rounding and for the difference between the simplified and full models.

1. Find the day number N of 15 August 2025.
2. Using Eq. (15), compute the declination on that date. Compare with NOAA's exact series, Eq. (6): why are the two not identical?
3. Using Eq. (16), compute the equation of time. Is a sundial fast or slow on that date, and by how much?
4. Using Eq. (25), compute the sunrise azimuth. Express your answer (i) in degrees from North, (ii) as an angle north or south of due east, (iii) as one of the sixteen compass names (N, NNE, NE, ...).
5. Using Eqs. (23) and (27), compute the sunrise time in IST.

6. Using Eq. (22), compute the Sun's azimuth at 09:00 IST on that date, and its altitude from Eq. (3).
7. A vertical pole 2 m tall stands at the site. Using the altitude from E6, find the length of its shadow at 09:00 IST.
8. At what latitude north does the Sun fail to rise at all on 21 December (using $z_0 = 90.833^\circ$ and $\delta = -23.44^\circ$)? Compare your answer with the latitude of the Arctic Circle (66.56° N) and explain the difference.
9. Show algebraically, from Eq. (25) and Eq. (26), that at any place the sunrise and sunset azimuths add up to 360° . Then explain why the actual numbers in Table Table 7 fail this rule by up to 0.2° .
10. **(Challenge)** Prove Eq. (22) directly from the two direction-cosine expressions, Eqs. (19) and (20), by dividing the first component by the second and simplifying.
11. **(Project)** Repeat E4–E6 for 21 June and for 21 December, then plot the three rising points on a plan view of the horizon like Fig. Fig. 5. Mark the position of due east. By how many degrees does the June rising point lie north of due east?

Answers

E1. $N = 31 + 28 + 31 + 30 + 31 + 30 + 31 + 15 = 227$ (2025 is not a leap year). Note that 15 August is the 227th day, so 148 days remain in the year.

E2. $\delta_0 = 23.44^\circ \sin[1.8(227 - 81)] = 23.44^\circ \sin 144.0^\circ = 13.78^\circ$ ($\pm 0.01^\circ$). NOAA's series gives 14.03° . The difference (0.25°) exists because the simplified formula assumes the Earth's orbit is a circle traversed at constant speed, while the real orbit is an ellipse (Kepler's second law); the difference is largest in September–October.

E3. $B = 1.8(227 - 81) = 144.4^\circ$; $EoT_0 = 9.87 \sin(288.8^\circ) - 7.53 \cos(144.4^\circ) - 1.5 \sin(144.4^\circ) = -9.34 + 6.12 - 0.87 = -4.10$ min. NOAA's series gives -4.55 min. A negative equation of time means *apparent* solar time is behind mean time, so a sundial reads about four minutes slow compared with the clock.

E4. $A_0 = 74.6^\circ$ (NOAA: 74.3°), i.e. 15.4° north of due east, compass direction ENE. The rising point is north of east because the declination is positive (summer in the northern hemisphere).

E5. $\cos H_0 = \frac{\cos 90.833^\circ}{\cos 23.17^\circ \cos 13.78^\circ} - \tan 23.17^\circ \tan 13.78^\circ = -0.01629 - 0.10501 = -0.12130$, so $H_0 = 96.97^\circ$ and sunrise is 6.46 hours before solar noon; with the corrections of Eq. (27) the clock time is 05:37 IST, in exact agreement with NOAA's algorithm.

E6. At 09:00 IST the hour angle is $H = -46.2^\circ$ and Eq. (22) gives $A = 93.7^\circ$ (NOAA: 93.3°) — the Sun is just north of due east, climbing. Its altitude is 45.4° .

E7. Shadow length $= h/\tan a = 2/\tan 45.4^\circ = 1.97$ m, pointing $93.7^\circ + 180^\circ = 273.7^\circ$, i.e. just north of due west.

E8. Solving $\cos H_0 = -1$ with $\delta = -23.44^\circ$ gives $\phi = 67.39^\circ$ N. The traditional Arctic Circle figure of 66.56° assumes the Sun's centre is on the mathematical horizon ($z = 90^\circ$) and ignores refraction;

because refraction lifts the Sun into view, the true “no sunrise” boundary lies about 0.8° further north — refraction is worth nearly 90 km of latitude.

E9. Sunset occurs at $-H_0$ with the same δ , and $\cos(360^\circ - A_0) = \cos A_0$, so the two azimuths are mirror images: $A_{\text{rise}} + A_{\text{set}} = 360^\circ$ exactly. In reality δ changes during the day by up to 0.2° (about 0.01° per hour), so the measured sums differ slightly — which is exactly the spread seen in the tables.

E10. Divide the first component of Eq. (20) by the second, use $\tan A = (\cos \delta \sin A)/(\cos \delta \cos A)$, divide numerator and denominator by $\cos \delta \cos \phi$, and use $\sin \phi / \cos \phi = \tan \phi$. The result is Eq. (21).

E11. 21 June: $A_0 = 63.96^\circ$ (26.0° north of due east); 21 December: $A_0 = 115.24^\circ$ (25.2° south of due east). The annual swing is 51.3° .

Conclusion

We set out to answer two questions a twelve-year-old asks and a twelfth-standard student should be able to answer: *where does the Sun rise, and where is it right now?* Starting from the solar-position equations published by NOAA — a verified source, quoted verbatim — and using only the trigonometry and inverse trigonometry of the NCERT syllabus, we obtained:

1. a single **general equation** for the Sun's azimuth,

$$A = 180^\circ + \tan^{-1} \left[\frac{\sin H}{\cos H \sin \phi - \tan \delta \cos \phi} \right],$$

which requires no morning/afternoon case analysis, is valid at all hours and seasons, and is algebraically equivalent to NOAA's published form;

2. a complete **sunrise formula** $A_0 = \cos^{-1}[(\sin \delta - \sin \phi \cos z_0)/(\cos \phi \sin z_0)]$, with $z_0 = 90.833^\circ$, and the matching sunrise-time formula;
3. a **verified simplification**: the textbook declination and equation-of-time formulas are the leading harmonics of NOAA's series (Table Table 4), and the resulting model reproduces the sun's rising point for all 365 days of 2025 with a maximum error of 1.16° (rms 0.57°) and the rising time to within 2.0 minutes, at a fraction of the arithmetic;
4. a complete **case study for a 22 km region** centred on 23.169753993380713° N, 82.36029982215409° E — Duman Hill, near Chirimiri, Khadganva tahsil, Manendragarh–Chirimiri–Bharatpur district, Chhattisgarh — showing that the Sun rises between 63.96° (21 June) and 115.24° (21 December), that this 51.3° swing is the whole seasonal story, and that inside the entire 22 km disc the rising direction varies by less than 0.09° and the rising time by less than 1.9 minutes, so that a single table serves the whole region;
5. an **independent check** against a public ephemeris service, which agrees with our computations to 0.14° and reproduces the live solar direction at the moment of writing;
6. reproducible **tables, code and exercises** (Appendices Appendix A and Appendix B) so that any reader, school or district office can regenerate every number for any place, date or year.

The deeper lesson is the one the project was framed around. We have not invented anything: the equations belong to NOAA, the geometry belongs to the textbooks of navigation, the constants are Kepler's and Newton's. What we have done is to take a professionally verified source, reduce it

honestly to the level of a Class-XII classroom, check the reduction numerically against the source itself, and then use it to answer a real question about a real place, exactly — to a tenth of a degree, which is finer than the eye can see. That, more than any formula, is what a school science project is for.

Appendix A. NOAA's azimuth routine, as published

The following listing is quoted verbatim (whitespace trimmed) from the NOAA Solar Calculator's JavaScript source, Ref. [2], the program that produced the reference values used throughout this paper. It is reproduced here as evidence of provenance: a reader can compare it line by line with Eqs. (5)–(11) and with our $\tan^{-1}$ form (Appendix Appendix C). NOAA publishes this code for public use.

```
function calcAzEl(T, localtime, latitude, longitude, zone) {

    var eqTime = calcEquationOfTime(T)
    var theta = calcSunDeclination(T)

    var solarTimeFix = eqTime + 4.0 * longitude - 60.0 * zone
    var earthRadVec = calcSunRadVector(T)
    var trueSolarTime = localtime + solarTimeFix
    while (trueSolarTime > 1440) {
        trueSolarTime -= 1440
    }
    var hourAngle = trueSolarTime / 4.0 - 180.0;
    if (hourAngle < -180) {
        hourAngle += 360.0
    }
    var haRad = degToRad(hourAngle)
    var csz = Math.sin(degToRad(latitude)) * Math.sin(degToRad(theta)) +
Math.cos(degToRad(latitude)) * Math.cos(degToRad(theta)) * Math.cos(haRad)
    if (csz > 1.0) {
        csz = 1.0
    } else if (csz < -1.0) {
        csz = -1.0
    }
    var zenith = radToDeg(Math.acos(csz))
    var azDenom = ( Math.cos(degToRad(latitude)) *
Math.sin(degToRad(zenith)) )
    if (Math.abs(azDenom) > 0.001) {
        var azRad = (( Math.sin(degToRad(latitude)) *
Math.cos(degToRad(zenith)) ) - Math.sin(degToRad(theta))) / azDenom
        if (Math.abs(azRad) > 1.0) {
            if (azRad < 0) {
                azRad = -1.0
            } else {
                azRad = 1.0
            }
        }
    }
    var azimuth = 180.0 - radToDeg(Math.acos(azRad))
    if (hourAngle > 0.0) {
        azimuth = -azimuth
    }
} else {
    if (latitude > 0.0) {
        var azimuth = 180.0
    }
}
```

```

    } else {
        var azimuth = 0.0
    }
}
if (azimuth < 0.0) {
    azimuth += 360.0
}
var exoatmElevation = 90.0 - zenith

// Atmospheric Refraction correction
var refractionCorrection = calcRefraction(exoatmElevation)

var solarZen = zenith - refractionCorrection;
var elevation = 90.0 - solarZen

return {"azimuth": azimuth, "elevation": elevation}
}

```

NOAA's `calcAzEL()`: the reference implementation of the solar azimuth.

Note the final operation: NOAA computes $180^\circ - \cos^{-1}(\cdot)$ and then negates the result if the hour angle is positive — the “mirror image” rule that our Eq. (22) absorbs automatically.

Appendix B. A complete program

The program below is self-contained: it contains every equation of Section 4–Section 5 and nothing else. It runs on any Python 3 installation, needs no libraries, and prints the numbers of the worked example of Section 6. Lines are commented with the equation numbers they implement.

```

# sun_azimuth.py -- "Where is the Sun right now?" (Class XII, Python 3)
# Site: 23.1698 N, 82.3603 E (IST = UTC+5:30). Angles in degrees.
from math import sin, cos, tan, asin, acos, atan2, radians, degrees

LAT, LON, TZ, LON_STD = 23.1698, 82.3603, 5.5, 82.5      # site and IST
meridian

def day_number(d, m, y):                                # 1 January = day 1
    cum = [0, 0, 31, 59, 90, 120, 151, 181, 212, 243, 273, 304, 334]
    leap = (y % 4 == 0 and y % 100 != 0) or y % 400 == 0
    return cum[m] + d + (1 if (m > 2 and leap) else 0)

def declination(n):                                     # Eq. (8): Cooper (1969)
    return 23.44 * sin(radians(360.0 / 365.0 * (n - 81)))

def equation_of_time(n):                                # Eq. (9): minutes
    B = radians(360.0 * (n - 81) / 364.0)
    return 9.87 * sin(2 * B) - 7.53 * cos(B) - 1.5 * sin(B)

def sun_azimuth(clock_hours, n):
    """Azimuth of the Sun (degrees clockwise from true North) and its
    altitude."""
    solar_time = clock_hours + equation_of_time(n) / 60.0 + (LON - LON_STD)
    / 15.0
    H = 15.0 * (solar_time - 12.0)                      # Eq. (10): hour angle,
    degrees

```

```

    d = radians(declination(n))          # declination
    p = radians(LAT)                     # latitude
    h = radians(H)
    A = 180.0 + degrees(atan2(sin(h), cos(h) * sin(p) - tan(d) * cos(p)))
# Eq. (13)
    z = degrees(acos(sin(p) * sin(d) + cos(p) * cos(d) * cos(h)))
# zenith angle
    return A % 360.0, 90.0 - z

def sunrise(clock_date=None, n=None):
    """Sunrise azimuth, Eq. (12), and sunrise clock time, Eq. (14)+EoT."""
    n = n or day_number(clock_date[0], clock_date[1], clock_date[2])
    d, p, z0 = radians(declination(n)), radians(LAT), radians(90.833)
    A = degrees(acos((sin(d) - sin(p) * cos(z0)) / (cos(p) * sin(z0))))
    H0 = degrees(acos(cos(z0) / (cos(p) * cos(d)) - tan(p) * tan(d)))
    t_solar = 12.0 - H0 / 15.0 #
sunrise, solar time
    t_clock = t_solar - equation_of_time(n) / 60.0 - (LON - LON_STD) / 15.0
    return A, t_clock

def hhmm(x):
    return"%02d:%02d"%(int(x),round(60*(x-int(x))))

DAY = 15
MONTH = 6
YEAR = 2026

n = day_number(DAY, MONTH, YEAR)

print("%02d/%02d/%d, site 23.1698 N 82.3603 E" % (DAY, MONTH, YEAR))
print(" declination      = %+.3f deg" % declination(n))
print(" equation of time = %+.2f min" % equation_of_time(n))

for t in (6, 8, 10, 12, 14, 16, 18):
    A, alt = sun_azimuth(t, n)
    print(" %02d:00 IST azimuth %7.3f deg altitude %6.2f deg" % (t, A, alt))

A, tr = sunrise((DAY, MONTH, YEAR))
print(" sunrise: %s IST, azimuth %.3f deg (i.e. %.3f deg south of due east)" % (hhmm(tr), A, A - 90))

```

A complete, dependency-free program for the Sun's azimuth at the study site.

Appendix C. Equivalence of the general equation and NOAA's printed form

NOAA's printed azimuth equation is

$$\cos(180^\circ - \theta) = -\frac{\sin\phi \cos z - \sin\delta}{\cos\phi \sin z} \quad (\text{Eq. (11)}).$$

Write $\cos(180^\circ - \theta) = -\cos\theta$ and use $\cos z = \sin\phi \sin\delta + \cos\phi \cos\delta \cos H$ (Eq. (3)). Then

$$\cos\theta = \frac{\sin\phi \cos z - \sin\delta}{\cos\phi \sin z}, \quad \sin z = \sqrt{1 - \cos^2 z},$$

and, after multiplying the numerator by $\cos\phi \cos\delta \cos H - \sin\phi \sin\delta + \cos z$ and simplifying with $\sin^2 z = 1 - \cos^2 z$, one obtains

$$\cos\theta = \frac{\cos\phi (\cos H \sin\phi - \tan\delta \cos\phi)}{\sqrt{(\cos H \sin\phi - \tan\delta \cos\phi)^2 + \sin^2 H}}, \quad \sin\theta = \frac{-\sin H}{\sqrt{(\cos H \sin\phi - \tan\delta \cos\phi)^2 + \sin^2 H}},$$

so that $\tan\theta = -\sin H / (\cos H \sin\phi - \tan\delta \cos\phi)$. The principal value $\theta = \tan^{-1}(\cdot)$ therefore reproduces NOAA's before-noon azimuth, and adding 180° — which the two-argument tangent does by itself when $\sin H > 0$ — reproduces the afternoon values, i.e.

$$A = 180^\circ + \tan^{-1} \left[\frac{\sin H}{\cos H \sin\phi - \tan\delta \cos\phi} \right] \equiv \text{Eq. (22)}.$$

The equivalence was also verified numerically over a grid of 61 000 points in (ϕ, δ, H) covering both hemispheres and all hour angles: the maximum difference between the two expressions was 2.7×10^{-5} degrees, i.e. rounding.

Appendix D. Data and code availability

The following files accompany this paper and regenerate every table and figure:

File	Contents
solar_azimuth.py	the complete engine: NOAA algorithm, NOAA printed sheet,
	the Class-XII model, verification, tables and the worked example
work/noaa_main.js	NOAA's published program, as downloaded (unmodified)
work/noaa_oracle.js	driver that executes NOAA's program to produce reference values
work/student_program.py	the student program of Appendix Appendix B
work/make_figures.py,	figure and table generators
work/make_tables.py	
data/annual_2025.csv	the annual table (47 dates) in machine-readable form
data/region_22km.csv	the region table over the 22 km disc
work/verification.json	all residuals behind Tables Table 10 and Table 11

- NOAA Global Monitoring Laboratory, *General Solar Position Calculations* (solar-equations sheet), <https://gml.noaa.gov/grad/solcalc/solareqns.PDF> (accessed 3 October 2026). The source of Eqs. (4)–(14).
- NOAA Global Monitoring Laboratory, *Solar Calculator* (source code), <https://gml.noaa.gov/grad/solcalc/> and <https://gml.noaa.gov/grad/solcalc/main.js> (accessed 3 October 2026). The reference algorithm of this paper; the azimuth routine is quoted in Appendix Appendix A.
- NOAA Global Monitoring Laboratory, *NOAA Solar Calculator* (web interface), <https://gml.noaa.gov/grad/solcalc/>.

- Time and Date AS, *Sunrise, sunset and sun position for 23°10'11" N, 82°21'37" E*, <https://www.timeanddate.com/sun/> (accessed 3 October 2026). Independent ephemeris used in Section 7.4.
- P. I. Cooper, “The absorption of radiation in solar stills”, *Solar Energy* **12**, 333–346 (1969). Source of the declination approximation, Eq. (15).
- CED Engineering, *Solar Energy Fundamentals* (course notes), <https://www.cedengineering.com>; states the three-term equation of time, Eq. (16), in the standard solar-engineering form.
- “Equation of Time”, technical note, <https://pcool.dyndns.org/index.php/equation-of-time/> (accessed 3 October 2026): the three-term series and the associated sunrise formulas.
- J. Meeus, *Astronomical Algorithms*, 2nd ed., Willmann-Bell (1998). The full-precision treatment behind the simplifications.
- NOAA National Centers for Environmental Information and British Geological Survey, *World Magnetic Model 2025*, <https://www.ncei.noaa.gov/products/world-magnetic-model>; magnetic declination at the study site used in Section 6.7.
- Open-Meteo, *Elevation API* (Copernicus DEM), <https://open-meteo.com/en/docs/elevation-api> (accessed 3 October 2026); elevation of the site and of the region.
- NCERT, *The Earth: Our Habitat*, Geography, Class VI, Chapter 3, “Motions of the Earth”.
- NCERT, *Practical Work in Geography, Part I*, Class XI, Chapter 3, “Latitude, Longitude and Time”.
- NCERT, *Fundamentals of Physical Geography*, Class XI, Chapter 9, “Solar Radiation, Heat Balance and Temperature”.
- NCERT, *Mathematics*, Class XI, Chapter 3, “Trigonometric Functions”.
- NCERT, *Mathematics, Part I*, Class XII, Chapter 2, “Inverse Trigonometric Functions”.